HUMAN COMPUTER INTERACTION

UNIT-II

DESIGN & SOFTWARE PROCESS

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Introduction to DESIGN

What is Design?

- Design involves:
  - – achieving goals within constraints and trade-off between these
  - – understanding the raw materials: computer and human
  - – accepting limitations of humans and of design.
- The design process has several stages and is iterative and never complete.
- Interaction starts with getting to know the users and their context:
  - – finding out who they are and what they are like . . .
  - – talking to them, watching them.

golden rule of design

understand your materials

- understand computers
  - limitations, capacities, tools, platforms
- understand people
  - psychological, social aspects
  - human error
- and their interaction …

To err is human

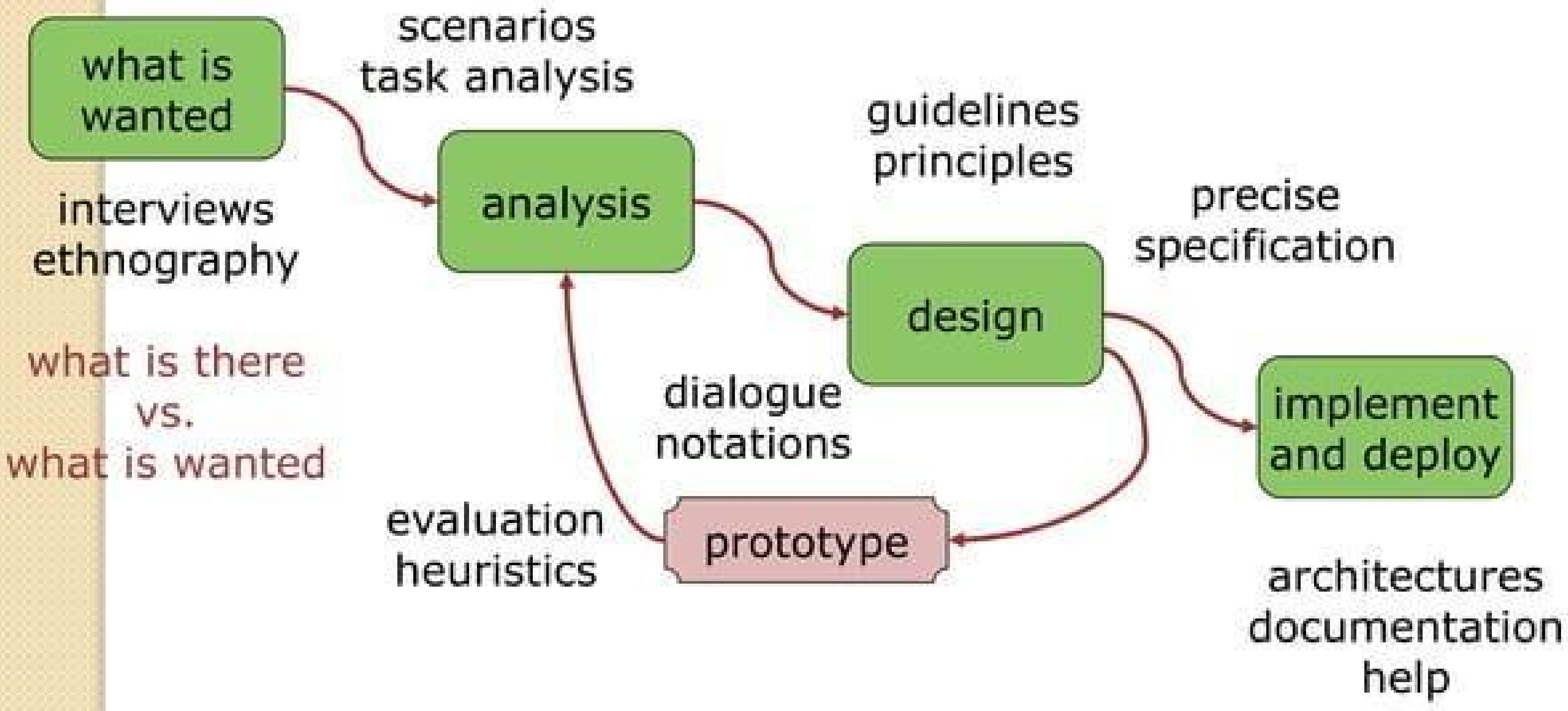
- nature of humans to make mistakes, and systems should be designed to reduce the likelihood of those mistakes and to minimize the consequences when mistakes happen.

The central message – the user

- This is the core of interaction design: put the user first, keep the user in the center and remember the user at the end.

process of Design

The process of Design



Steps …

- requirements
  - what is there and what is wanted …
- analysis
  - ordering and understanding
- design
  - what to do and how to decide
- iteration and prototyping
  - getting it right … and finding what is really needed!
- implementation and deployment
  - making it and getting it out there

USER FOCUS

- know your users
  - who are they?
    - Are they young or old, experienced computer users or novices?
  - probably <u>not</u> like you!
    - When designing a system it is easy to design it as if you were the main user: you assume your own interests and abilities. So often you hear a designer say 'but it's obvious what to do'.
  - talk to them
    - This can take many forms: structured interviews about their job or life,
    - open-ended discussions, or bringing the potential users fully into the design process.
    - The last of these is called *participatory design*
  - watch them
- use your imagination

Scenarios

Scenarios

- Scenarios are stories for design: rich stories of interaction.
- simplest design representation, but one of the most flexible and powerful.
- Scenarios can be used to:
  - communicate with others
  - validate other models
  - understand dynamics

- *Scenarios are a resource that can be used and reused throughout the design process: helping us see what is wanted, suggesting how users will deal with the potential design, checking that proposed implementations will work, and generating test cases for final evaluation.*

NAVIGATION DESIGN

NAVIGATION DESIGN

Different levels of interaction:

- **Widgets :**The appropriate choice of widgets and wording in menus and buttons will help you know how to use them for a particular selection or action.
- **Screens or windows** You need to find things on the screen, understand the logical grouping of buttons.
- **Navigation within the application** You need to be able to understand what will happen when a button is pressed, to understand where you are in the interaction.
- **Environment** The word processor has to read documents from disk, perhaps some are on remote networks. You swap between applications, perhaps cut and paste.

NAVIGATION DESIGN

2 main kinds of issue:

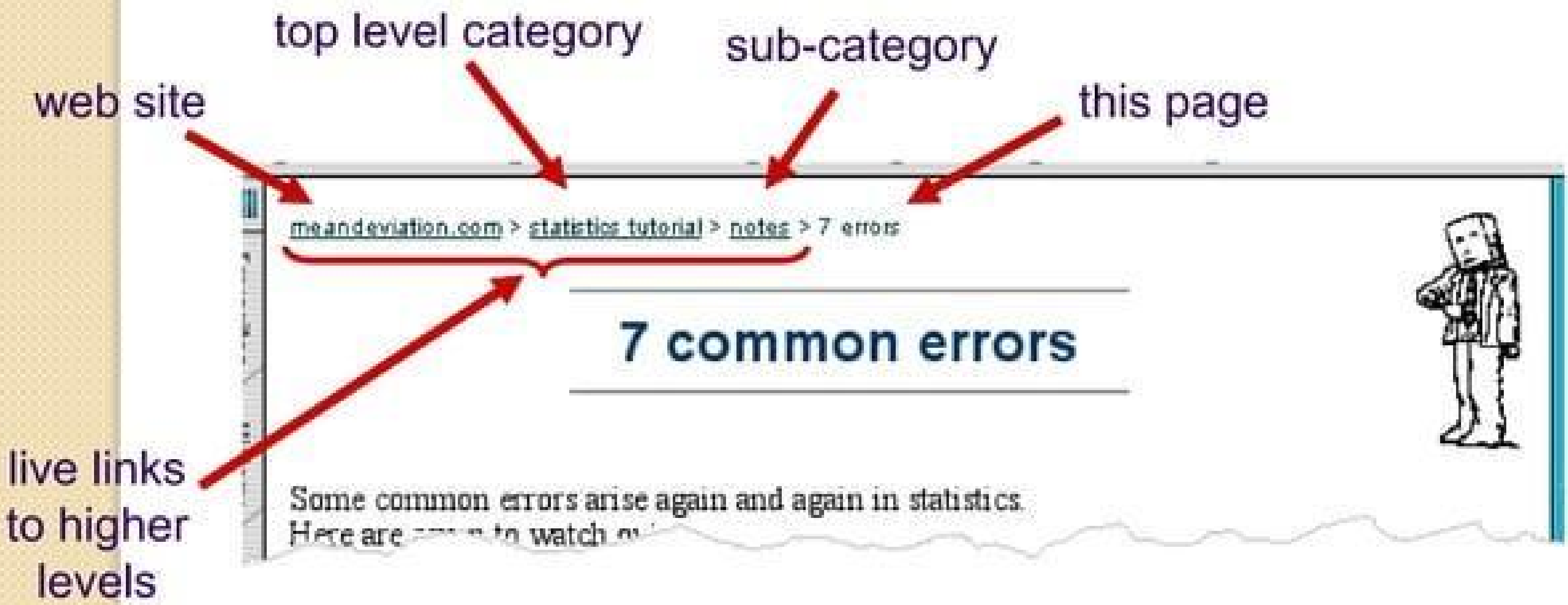
- local structure
  - – looking from one screen or page out
- global structure
  - – structure of site, movement between screens.

local structure

- Much of interaction involves goal-seeking behavior.
- To do this goal seeking, each state of the system or each screen needs to give the user enough knowledge of what to do to get closer to their goal
- four things to look:
  - knowing where you are
  - knowing what you can do
  - knowing where you are going – or what will happen
  - knowing where you've been – or what you've done.

where you are – breadcrumbs

shows path through web site hierarchy

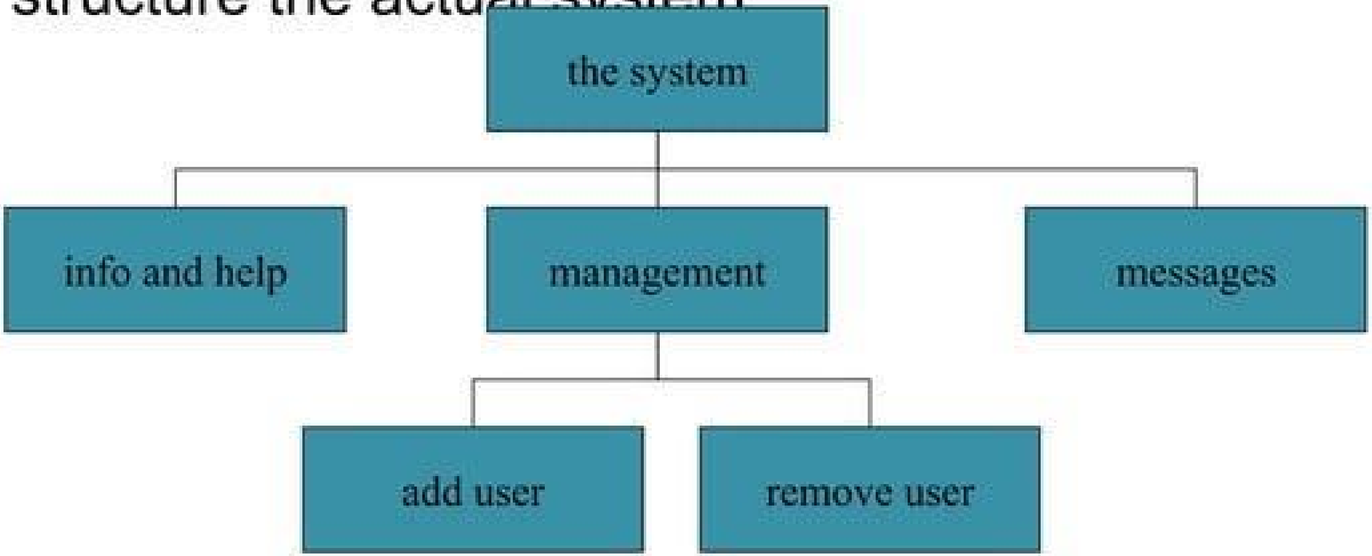


- *what you can do* – *what can be pressed or clicked to go* somewhere or do something.
- On the web the standard underlined links make it clear which text is clickable and which is not.
- To improve the appearance of the page many sites change the color of links and may remove the underline

- where you are going when you click a button or what will happen.
- *what you've done* - need to know everything about the internal state of the system and things outside, like the contents of files, networked devices, etc., that could affect it

Global structure – hierarchical organization

- One way to organize a system is in some form of hierarchy.
- organized along functional boundaries, by roles, user type, or some more esoteric breakdown such as modules.
- help during design, but can also be used to structure the actual system



- deep hierarchies are difficult to navigate
- better to have broad top-level categories, or to present several levels of menu on one screen or web page
- Eg: ,
  - if the user wants to look up information on a particular city, an alphabetical list of all city names would be fast, but for other purposes a list by region would be more appropriate.

Global structure – dialog

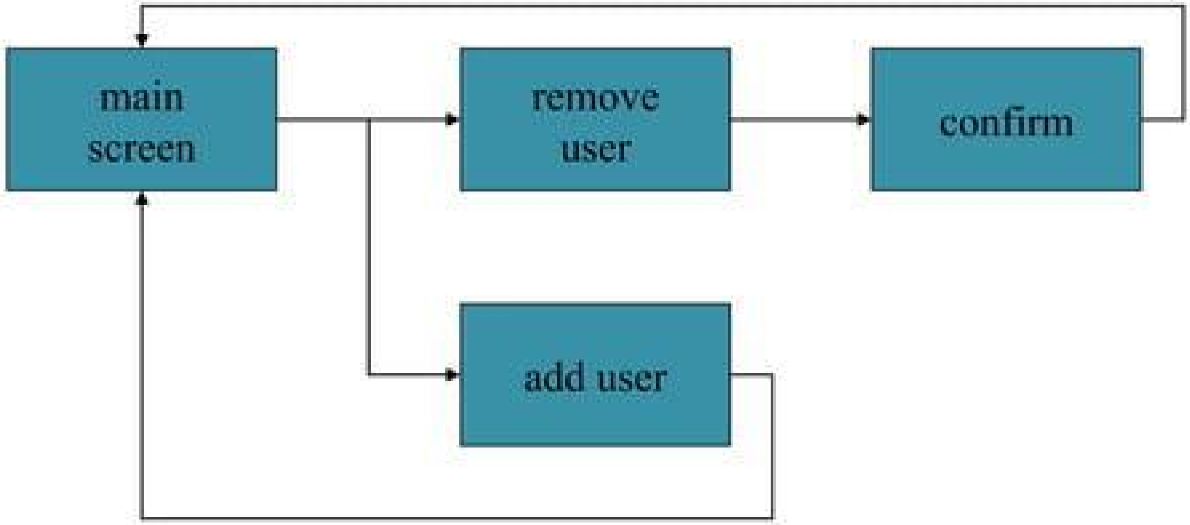
- In HCI, 'dialog' is used to refer to the pattern of interactions between the user and a system.

Minister: do you *name* take this woman …
Man: I do
Minister: do you *name* take this man …
Woman: I do
Minister: I now pronounce you man and wife

- marriage service
  - general flow, generic – blanks for names
  - pattern of interaction between people
- computer dialogue
  - pattern of interaction between users and system
  - but details differ each time.

- To describe a full system we need to take into account different paths through a system and loops where the system returns to the same screen.
- Simple way is Network diagram

network diagrams



show what leads to what
show what happens when
include branches and loops
be more task oriented than a
hierarchy

SCREEN DESIGN AND LAYOUT

SCREEN DESIGN AND LAYOUT

Basic principles

- ask
  - what is the user doing?
- think
  - what information, comparisons, order
- design
  - form follows function

Tools for layout

- grouping of items
- order of items
- decoration - fonts, boxes etc.
- alignment of items
- white space between items

grouping and structure

logically together $\Rightarrow$ physically together

Billing details:
Name
Address: …
Credit card no

Delivery details:
Name
Address: …
Delivery time

Order details:

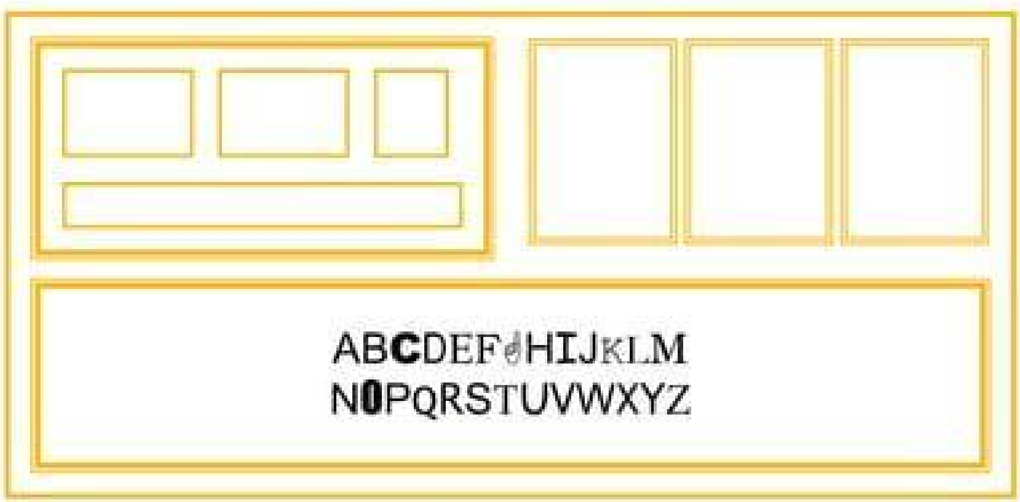
item	quantity	cost/item	cost
`size 10 screws (boxes)`	`7`	`3.71`	`25.97`
……	…	…	…

order of groups and items

- think! - what is natural order
- should match screen order!
- For data entry forms or dialog boxes we should also set up the order in which the tab key moves between fields.

decoration

- use boxes to group logical items
- decorative features like font style, and text or background colors can be used to emphasize groupings
- but not too many!!



alignment - text

- you read from left to right (English and European)

⇒ align left hand side

boring but readable!

Willy Wonka and the Chocolate Factory
Winston Churchill - A Biography
Wizard of Oz
Xena - Warrior Princess

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Winston Churchill - A Biography
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fine for special effects but hard to scan

alignment - names

- Usually scanning for surnames

  ⇒ make it easy!

Alan Dix
Janet Finlay
Gregory Abowd
Russell Beale

Alan	Dix
Janet	Finlay
Gregory	Abowd
Russell	Beale

Dix , Alan
Finlay, Janet
Abowd, Gregory
Beale, Russell

alignment - numbers

think purpose!

which is biggest?

532.56
179.3
256.317
15
73.948
1035
3.142
497.6256

alignment - numbers

visually:

long number = big number

align decimal points

or right align integers

627.865
1.005763
382.583
2502.56
432.935
2.0175
652.87
56.34

multiple columns

sherbert	75
toffee	120
chocolate	35
fruit gums	27
coconut dreams	85

(i)

sherbert	75
toffee	120
chocolate	35
fruit gums	27
coconut dreams	85

(ii)

sherbert	75
toffee	120
chocolate	35
fruit gums	27
coconut dreams	85

(iii)

sherbert	75
toffee	120
chocolate	35
fruit gums	27
coconut dreams	85

(iv)

Figure 5.11 Managing multiple columns

White space

- In typography the space between the letters is called the counter

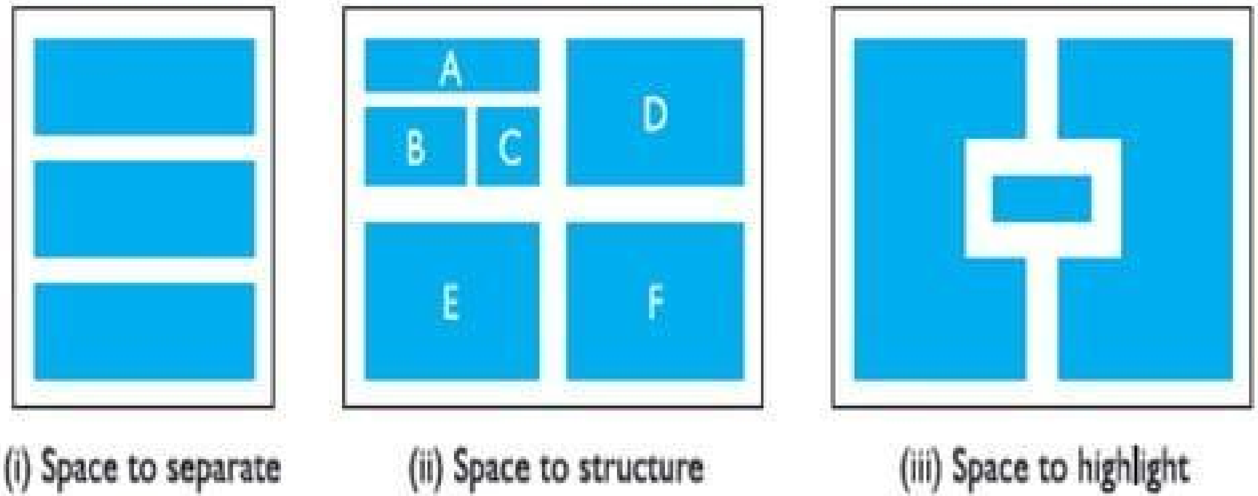
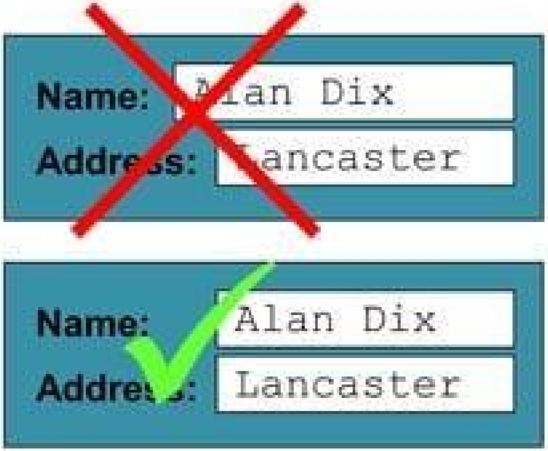


Figure 5.12 Using white space in layout

User action and control entering information

- forms, dialogue boxes
  - presentation + data input
  - similar layout issues
  - alignment - N.B. different label lengths
- logical layout
  - use task analysis (ch15)
  - groupings
  - natural order for entering information
    - top-bottom, left-right (depending on culture)
    - set tab order for keyboard entry



N.B. see extra slides for widget choice

knowing what to do

- what is active what is passive
  - where do you click
  - where do you type
- consistent style helps
  - e.g. web underlined links
- labels and icons
  - standards for common actions
  - language – bold = current state or action

affordances

- The psychological idea of *affordance says that things may*

suggest by their shape and other attributes what you can do to them

- for physical objects
  - shape and size suggest actions
    - pick up, twist, throw
  - also cultural – buttons 'afford' pushing
- for screen objects
  - button–like object 'affords' mouse click
  - physical-like objects suggest use
- culture of computer use
  - icons 'afford' clicking
  - or even double clicking … not like real buttons!

Appropriate appearance presenting information

- The way of presenting information on screen depends on the kind of information: text, numbers, maps, tables; on the technology available to present it
- Eg: For more complex numerical data, may use scatter graphs, histograms or 3D surfaces; for hierarchical diagrams et

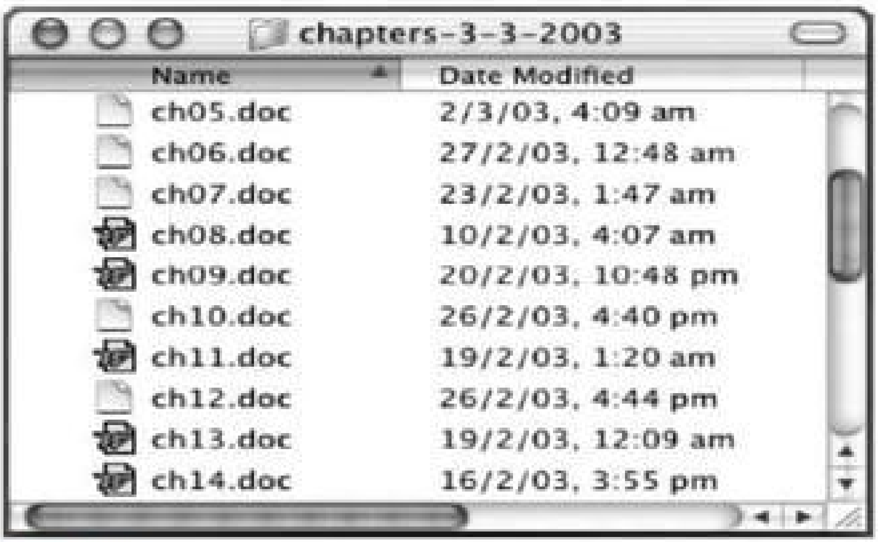


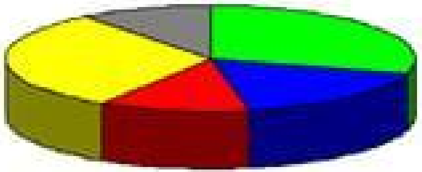
Figure 5.13 Alphabetic file listing. Screen shot reprinted by permission from Apple Computer, Inc.

Aesthetics and utility

- an interface should be aesthetically pleasing
- good graphic design and attractive displays can increase users' satisfaction and thus improve productivity
- The conflict between aesthetics and utility can also be seen in many 'welldesigned' posters and multimedia systems
- beauty and utility may conflict
  - mixed up visual styles ⇒ easy to distinguish
  - clean design – little differentiation ⇒ confusing
  - backgrounds behind text
    … good to look at, but hard to read

colour and 3D

- both often used very badly!
- colour
  - older monitors limited palette
  - colour over used because 'it is there'
  - beware colour blind!
  - use sparingly to reinforce other information
- 3D effects
  - good for physical information and some graphs
  - but if over used …
    e.g. text in perspective!! 3D pie charts



bad use of colour

- over use - without very good reason (e.g. kids' site)

- over use - without very good reason (e.g. kids' site)
- colour blindness
- poor use of contrast
- do adjust your set!
  - adjust your monitor to greys only
  - can you still read your screen?

across countries and cultures

- localisation & internationalisation
  - changing interfaces for particular cultures/languages
- globalisation
  - try to choose symbols etc. that work everywhere

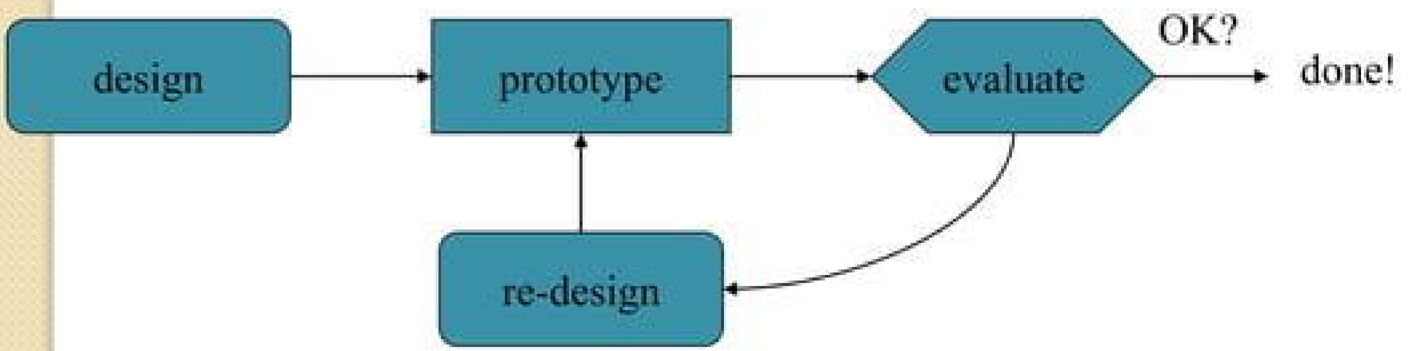
- deeper issues
  - cultural assumptions and values
  - meanings of symbols

    e.g tick and cross … +ve and -ve in some cultures
    … but … mean the same thing (mark this) in others

ITERATION AND PROTOTYPING

prototyping

- you never get it right first time
- if at first you don't succeed …



- Evaluation, intended to improve designs, is called *formative evaluation*.
- *Summative evaluation, is performed at the end to* verify whether the product is good enough.
- The result of evaluating the system will usually be a list of faults or problems followed by a redesign exercise, which is then prototyped, evaluated . . .
- iteration and prototyping are the universally accepted ‘best practice’ approach for interaction design

For prototyping methods to work:

- 1. To understand what is wrong and how to improve.
- 2. A good start point.

- cannot iterate the design unless you know what must be done to improve it.
- The second, is needed to avoid local maxima.
- A really good designer might guess a good initial design based on experience and judgment

Thank you